

# ASSIGNMET-2

(Manoj Sanagapalli)

**AI-Powered Customer Support Automation System**

**GitHub Repo Link:**

<https://github.com/Manoj-Sanagapalli20/Customer-Support-Chatbot>

**Readme.md:**

<https://github.com/Manoj-Sanagapalli20/Customer-Support-Chatbot/blob/main/README.md>

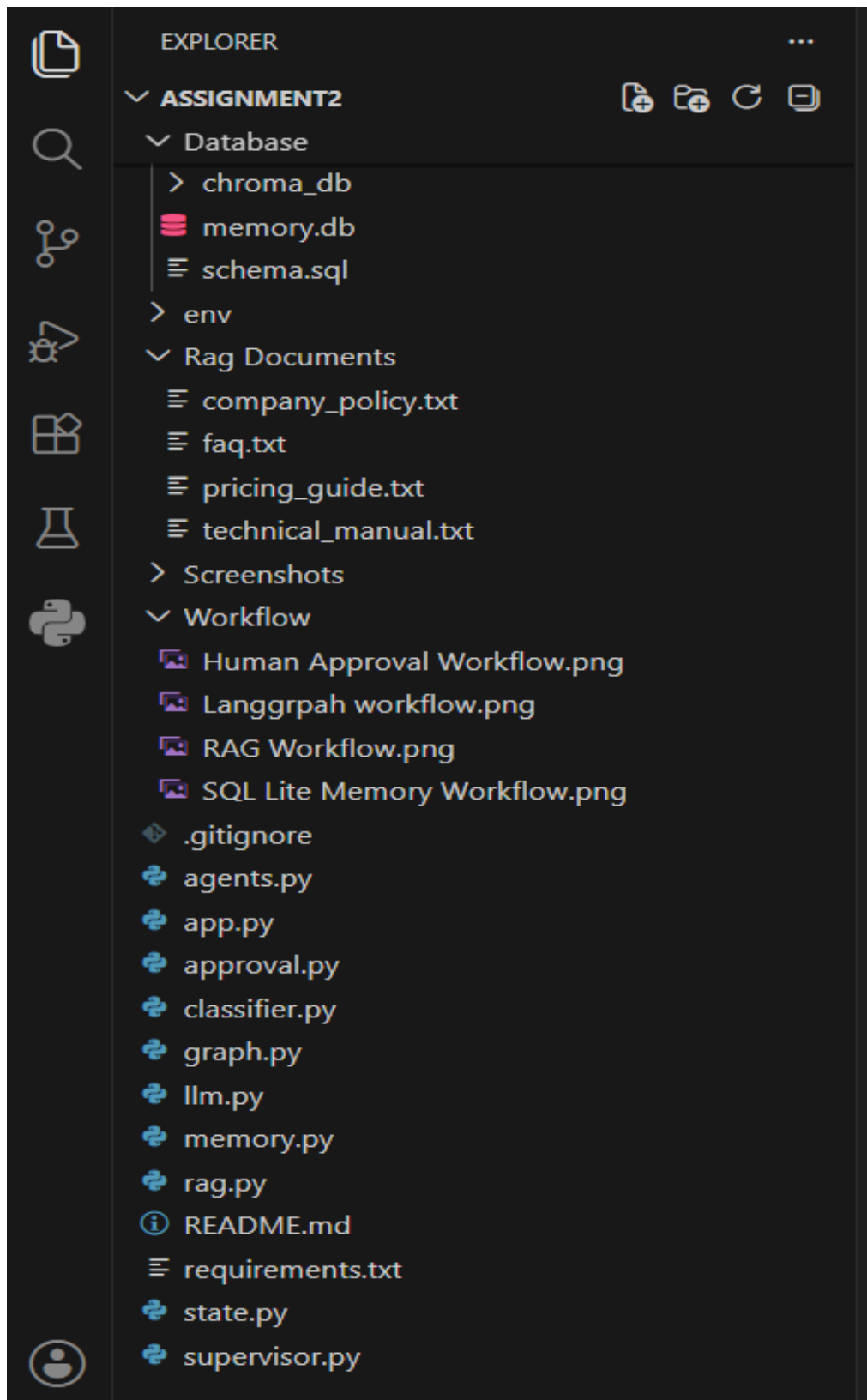
**DataBase Link:**

<https://github.com/Manoj-Sanagapalli20/Customer-Support-Chatbot/blob/main/Database/memory.db>

**Source Code Link (Zip Download):**

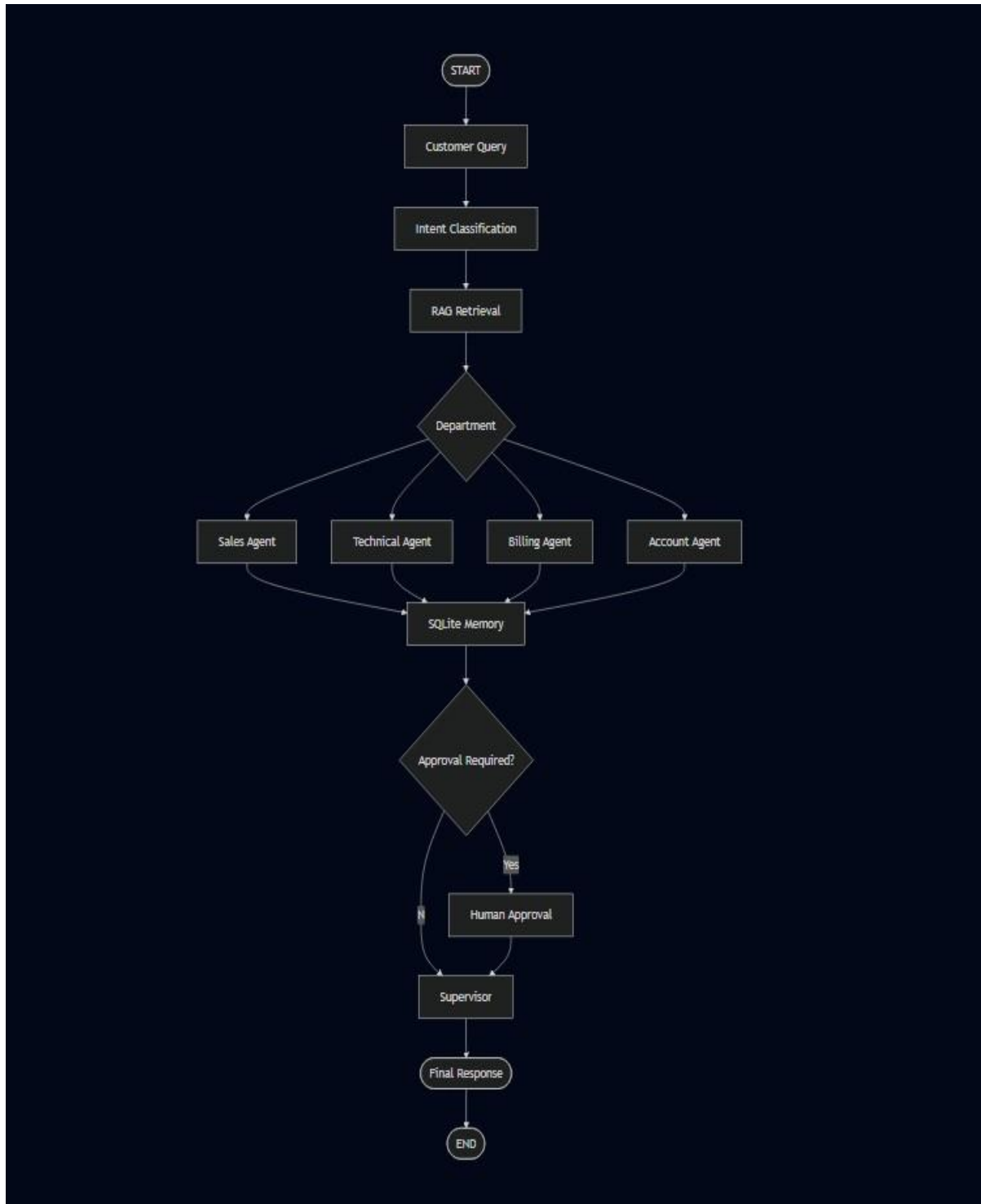
<https://github.com/Manoj-Sanagapalli20/Customer-Support-Chatbot/archive/refs/heads/main.zip>

## File Structure

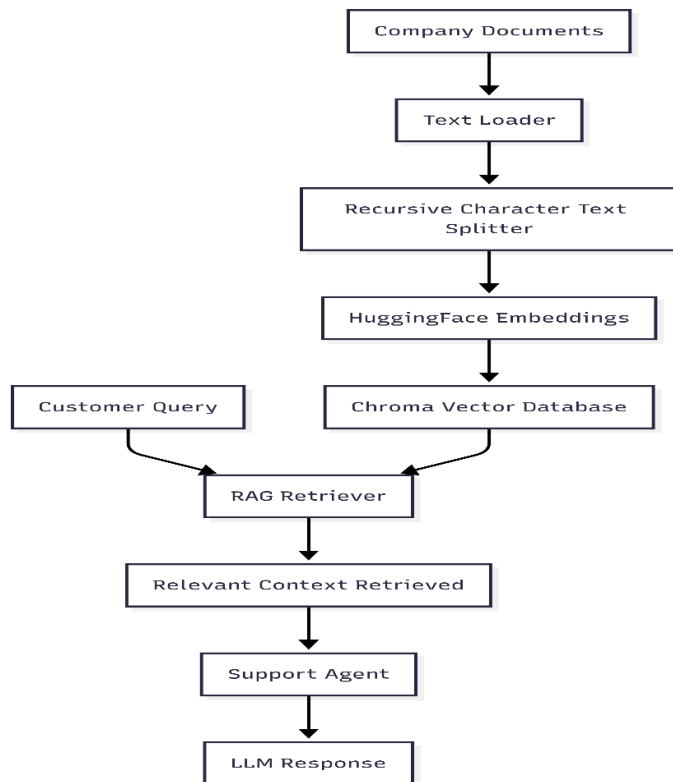


## Work Flows (Task 1)

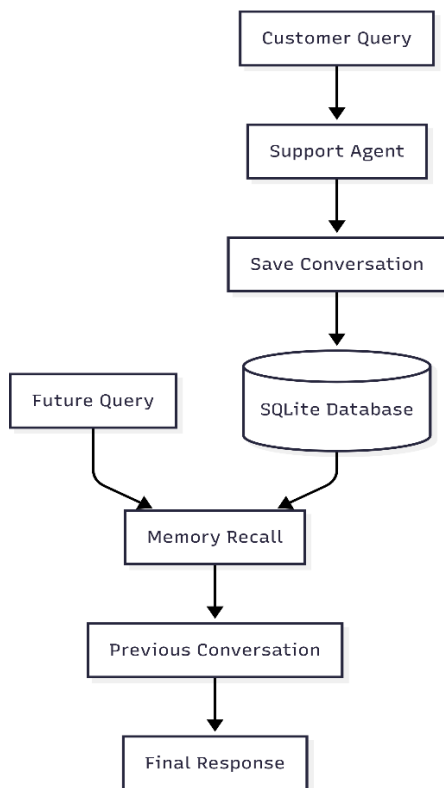
### LangGraph WorkFlow



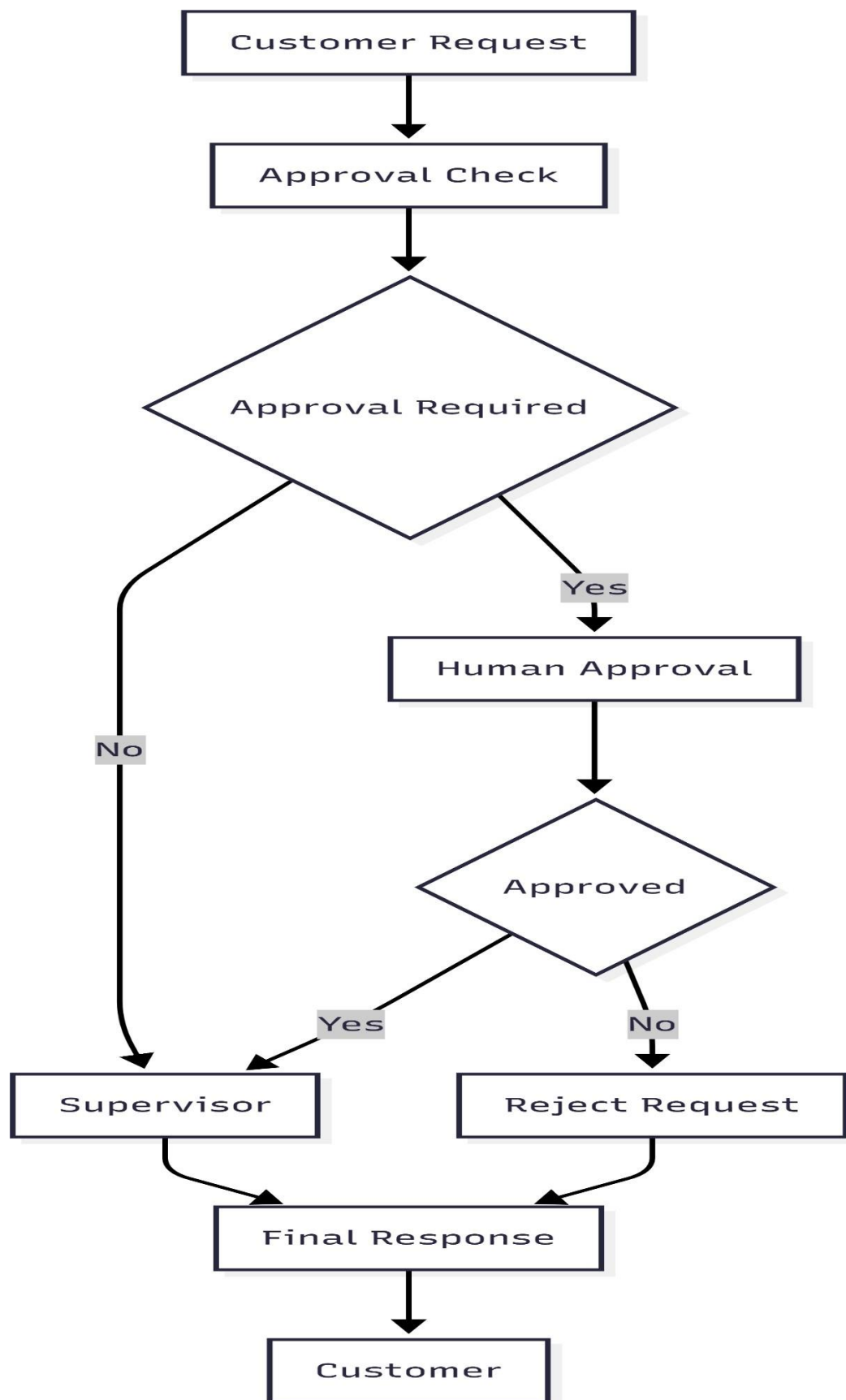
### RAG WorkFlow



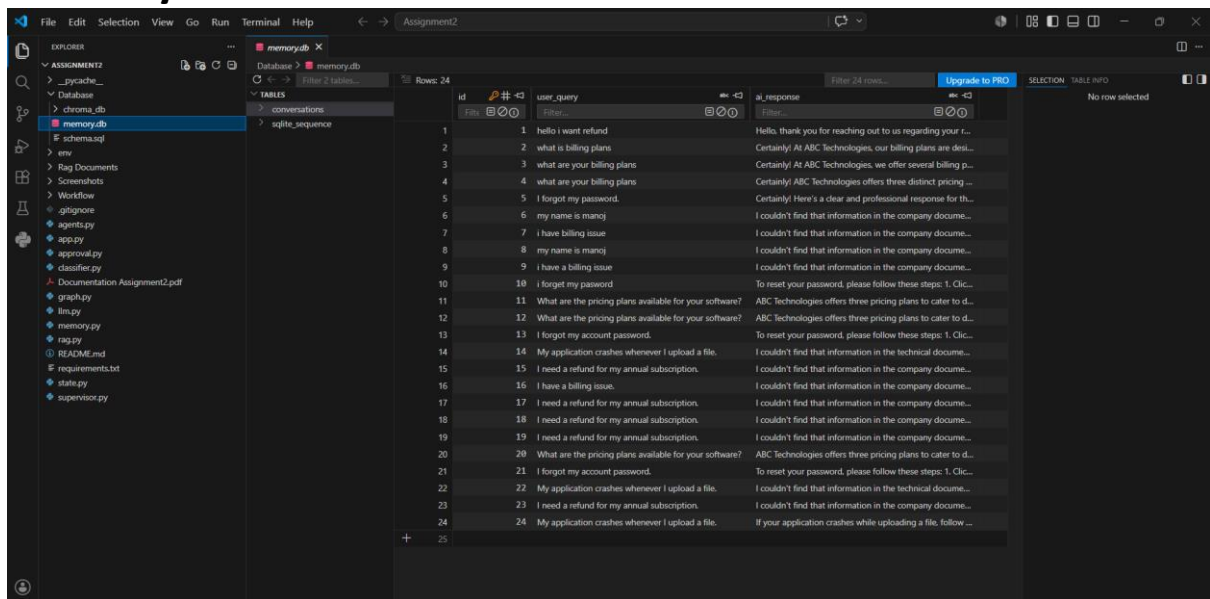
## SQL Lite WorkFlow



## Human Approval WorkFlow



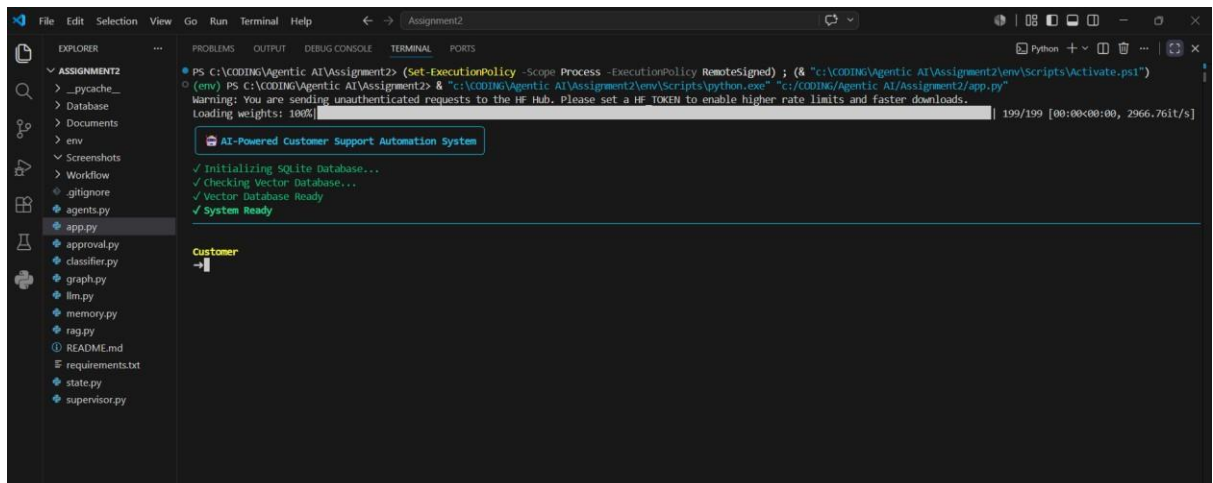
# Memory DB



| id | user_query                                              | ai_response                                                    |
|----|---------------------------------------------------------|----------------------------------------------------------------|
| 1  | hello i want refund                                     | Hello, thank you for reaching out to us regarding your r...    |
| 2  | what is billing plans                                   | Certainly! At ABC Technologies, our billing plans are desi...  |
| 3  | what are your billing plans                             | Certainly! At ABC Technologies, we offer several billing p...  |
| 4  | what are your billing plans                             | Certainly! ABC Technologies offers three distinct pricing ...  |
| 5  | I forgot my password.                                   | Certainly! Here's a clear and professional response for th...  |
| 6  | my name is manoj                                        | I couldn't find that information in the company docume...      |
| 7  | I have billing issue                                    | I couldn't find that information in the company docume...      |
| 8  | my name is manoj                                        | I couldn't find that information in the company docume...      |
| 9  | I have a billing issue                                  | I couldn't find that information in the company docume...      |
| 10 | I forgot my password                                    | To reset your password, please follow these steps: 1. Clic...  |
| 11 | What are the pricing plans available for your software? | ABC Technologies offers three pricing plans to cater to d...   |
| 12 | What are the pricing plans available for your software? | ABC Technologies offers three pricing plans to cater to d...   |
| 13 | I forgot my account password.                           | To reset your password, please follow these steps: 1. Clic...  |
| 14 | My application crashes whenever I upload a file.        | I couldn't find that information in the technical docume...    |
| 15 | I need a refund for my annual subscription.             | I couldn't find that information in the company docume...      |
| 16 | I have a billing issue.                                 | I couldn't find that information in the company docume...      |
| 17 | I need a refund for my annual subscription.             | I couldn't find that information in the company docume...      |
| 18 | I need a refund for my annual subscription.             | I couldn't find that information in the company docume...      |
| 19 | I need a refund for my annual subscription.             | I couldn't find that information in the company docume...      |
| 20 | What are the pricing plans available for your software? | ABC Technologies offers three pricing plans to cater to d...   |
| 21 | I forgot my account password.                           | To reset your password, please follow these steps: 1. Clic...  |
| 22 | My application crashes whenever I upload a file.        | I couldn't find that information in the technical docume...    |
| 23 | I need a refund for my annual subscription.             | I couldn't find that information in the company docume...      |
| 24 | My application crashes whenever I upload a file.        | If your application crashes while uploading a file, follow ... |

## OUTPUT SCREENSHOTS:

## Teminal



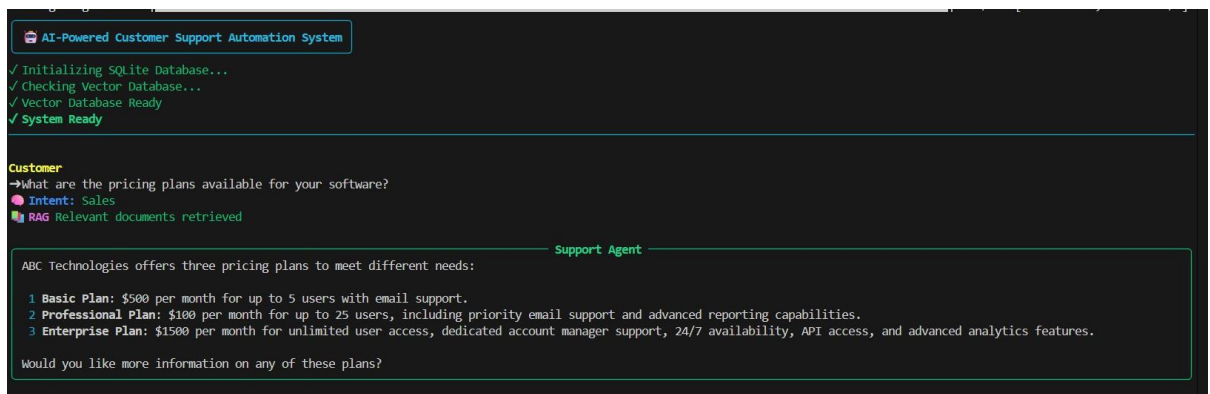
```
PS C:\CODING\Agentic AI\Assignment2> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned); (& "c:\CODING\Agentic AI\Assignment2\env\Scripts\Activate.ps1")
(env) PS C:\CODING\Agentic AI\Assignment2> & "c:\CODING\Agentic AI\Assignment2\env\Scripts\python.exe" "c:\CODING\Agentic AI\Assignment2\app.py"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 199/199 [00:00:00:00, 2966.76it/s]

AI-Powered Customer Support Automation System

✓ Initializing SQLite Database...
✓ Checking Vector Database...
✓ Vector Database Ready
✓ System Ready

Customer
→ What are the pricing plans available for your software?
Intent: Sales
RAG Relevant documents retrieved
```

## Query 1



```
AI-Powered Customer Support Automation System

✓ Initializing SQLite Database...
✓ Checking Vector Database...
✓ Vector Database Ready
✓ System Ready

Customer
→ What are the pricing plans available for your software?
Intent: Sales
RAG Relevant documents retrieved

Support Agent

ABC Technologies offers three pricing plans to meet different needs:

1 Basic Plan: $500 per month for up to 5 users with email support.
2 Professional Plan: $100 per month for up to 25 users, including priority email support and advanced reporting capabilities.
3 Enterprise Plan: $1500 per month for unlimited user access, dedicated account manager support, 24/7 availability, API access, and advanced analytics features.

Would you like more information on any of these plans?
```

## Query 2

Customer

→I forgot my account password.

● Intent: Account

■ RAG Relevant documents retrieved

Support Agent

To reset your password, please follow these steps:

- 1 Click on "Forgot Password".
- 2 Enter your registered email address.
- 3 You will receive an email with instructions to create a new password.
- 4 Create a new password using the provided link in the email.
- 5 Log in using your new password.

If you continue to experience issues, please verify your email address and ensure that your internet connection is stable. Clearing your browser cache or trying another browser might also help resolve any temporary issues. If these steps do not work, contact our technical support for further assistance.

## Query 3

Customer

→My application crashes whenever I upload a file.

● Intent: Technical

■ RAG Relevant documents retrieved

Support Agent

If your application crashes while uploading a file, follow these steps:

- 1 Ensure the file size is within the supported limit as specified in the technical documentation.
- 2 Verify that the file format you are trying to upload is supported by checking the list of supported formats provided in the documentation.
- 3 Restart the application to clear any temporary issues.
- 4 Update to the latest version of the application, as newer versions often include bug fixes and improvements for file upload functionality.
- 5 If the issue still persists after following these steps, please contact technical support for further assistance.

I couldn't find that information in the technical documentation.

## Query 4

Customer

→I need a refund for my annual subscription.

● Intent: Billing

■ RAG Relevant documents retrieved

⚠ Human Approval Required

I need a refund for my annual subscription.

Customer Request:

I need a refund for my annual subscription.

Approve request? (yes/no): yes

Support Agent

Thank you for reaching out. Your request for a refund on your annual subscription has been approved by our support team.

## Query 5

Customer

→What was my previous support issue?

● Intent: Memory

■ RAG Relevant documents retrieved

● Memory Previous conversation found

Support Agent

Your previous support issue was related to a refund request.

# **Manoj Sanagapalli**

## **Assignment 2**